

KALINGAPATNAM, India

WMO: 431050

Lat: 18.333N

Lon: 84.133E

Elev: 20

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.4	64.2	54.2	62.5	70.2	56.4	67.8	70.0	15.7	81.5	14.3	80.6	1.7	N/A	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.7	94.5	83.2	93.2	83.0	92.1	82.7	85.4	91.0	84.9	90.4	84.4	89.8	9.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
84.1	178.8	88.7	83.5	175.4	88.5	82.9	172.0	88.3	50.1	91.8	49.5	90.5	48.8	90.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	16.8	14.1	DB	57.9	100.5	1.8	3.9	56.6	103.3	55.6	105.6	54.6	107.8	53.3	110.7
			WB	55.1	85.9	1.6	0.2	54.0	86.1	53.0	86.3	52.1	86.4	51.0	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	73.8	77.9	82.9	85.2	86.9	86.1	83.9	84.3	84.1	82.5	78.1	74.3
	DBStd	4.94	2.54	2.65	1.96	1.80	2.45	3.03	2.20	2.03	1.92	2.49	2.40	2.84
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11558	737	781	1019	1056	1145	1083	1050	1062	1022	1006	843	754
	CDD65	6083	272	361	554	606	680	633	585	597	572	541	393	289
	CDH74	68410	1366	2691	6099	7861	9547	8562	7145	7324	6945	6033	3221	1616
	CDH80	27981	172	766	2254	3688	5182	4319	2800	2980	2754	2131	723	212
Wind	WSAvg	4.8	3.5	4.3	6.1	8.1	7.3	5.2	5.2	4.0	3.4	3.0	3.7	3.7
Precipitation	PrecAvg	61.3	0.4	0.6	0.9	1.9	4.3	7.7	12.2	12.4	10.8	7.4	2.3	0.6
	PrecMax	85.7	2.0	2.8	4.2	3.6	14.1	13.9	16.1	18.7	22.2	20.8	7.8	4.6
	PrecMin	47.7	0.0	0.0	0.1	0.4	1.2	3.5	7.0	8.1	6.2	1.6	0.0	0.0
	PrecStd	8.8	0.5	0.9	1.0	0.9	2.8	2.6	2.6	2.6	3.1	4.5	2.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.5	91.3	92.8	93.6	96.4	99.6	92.7	93.5	92.9	92.8	88.1	86.4
		MCWB	74.0	72.0	76.6	81.3	83.5	82.7	83.1	83.2	83.2	80.2	77.2	75.0
	2%	DB	83.6	88.3	90.9	92.4	94.3	94.9	90.9	91.4	91.4	90.8	86.4	83.8
		MCWB	73.1	74.6	78.3	81.7	83.7	84.4	82.8	83.0	82.9	80.0	76.2	73.6
	5%	DB	81.9	86.6	89.7	91.3	93.1	93.1	89.6	89.9	90.0	89.3	85.1	82.3
		MCWB	72.2	74.6	78.7	81.5	83.8	83.9	82.4	82.5	82.5	79.8	75.7	72.6
	10%	DB	80.3	84.7	88.4	90.2	91.9	91.6	88.3	88.7	88.8	87.7	83.9	80.8
		MCWB	71.5	74.3	78.7	81.3	83.6	83.5	82.1	82.1	82.1	79.7	75.3	72.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	78.9	81.8	83.9	86.2	85.6	84.5	84.6	84.5	83.4	80.4	77.7
		MCDB	82.3	86.2	88.9	90.7	91.3	91.9	89.8	90.6	90.8	88.7	85.5	83.0
	2%	WB	75.4	77.7	80.9	83.2	85.5	85.3	83.6	83.7	83.6	82.4	79.1	76.4
		MCDB	80.7	84.5	87.5	89.7	90.6	91.4	88.6	89.2	89.4	87.8	83.7	81.5
	5%	WB	74.0	76.8	80.3	82.7	85.1	84.8	82.9	83.0	82.9	81.7	78.2	75.0
		MCDB	79.5	83.4	86.6	88.9	90.2	90.7	87.9	88.2	88.3	86.8	82.6	79.9
	10%	WB	72.8	75.8	79.7	82.2	84.6	84.2	82.4	82.5	82.4	80.9	77.2	73.7
		MCDB	78.5	82.3	85.8	88.2	89.7	89.8	87.3	87.4	87.6	85.7	81.6	78.7
Mean Daily Temperature Range	5% DB	MDBR	15.3	16.2	12.9	11.2	10.7	9.5	7.9	8.7	9.2	10.6	12.7	14.2
		MCDBR	16.6	18.1	14.2	12.1	11.6	11.7	10.2	10.8	11.3	12.9	14.1	15.1
		MCWBR	8.2	7.5	5.2	4.7	5.1	5.0	4.6	4.8	5.1	4.8	6.3	6.8
	5% WB	MCDBR	14.1	14.8	11.8	10.6	9.6	10.7	9.2	9.9	10.4	10.6	10.9	12.4
		MCWBR	7.9	7.1	5.0	4.8	4.9	5.2	4.7	5.1	5.2	5.1	6.0	7.1
Clear-Sky Solar Irradiance	taub		0.526	0.521	0.599	0.600	0.648	0.659	0.580	0.554	0.554	0.550	0.536	0.552
	taud		1.943	1.983	1.833	1.882	1.791	1.774	1.970	2.048	2.044	2.028	1.984	1.896
	Ebn at Noon		238	249	234	235	221	217	235	243	242	238	234	225
	Edh at Noon		56	57	68	65	71	71	59	55	54	54	54	57
All-Sky Solar Radiation	RadAvg		1474	1762	1936	2091	1931	1454	1315	1351	1431	1484	1420	1347
	RadStd		60	70	90	63	88	157	151	118	113	142	121	91

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.54	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page